MUSSA MESHACK THOMAS

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PROFESSIONAL SUMMARY

Data Science undergraduate and Machine Learning Engineer with practical experience in healthcare data analytics, AI system development, and intelligent web-based solutions. Demonstrated capability in building end-to-end AI systems including Chabot, recommendation tools, and tracking platforms. Experience spans hospital data environments, university innovation labs, and national-level hackathons. Strong focus on applied artificial intelligence, problem-solving, and scalable digital solutions within academic and public sector contexts.

TECHNICAL SKILLS AND STRENGTHS

- Programming & Tools: Python, Data Visualization Libraries, ML Frameworks
- Machine Learning: Classification, Regression, Model Evaluation
- AI Development: NLP-based Chatbots, Generative AI Systems
- Web Development: Frontend–Backend Integration for AI Systems
- Data Analytics: Cleaning, Transformation, Statistical Interpretation
- Research, Experimentation & Model Optimization

PROJECTS

- Successfully built and deployed MUST AI Chatbot serving a university-wide academic community.
- Developed an AI-powered postal packet tracker prototype recognized among top finalists in a national hackathon hosted by Tanzania Communications Regulatory Authority.
- Currently architecting an AI-driven digital library assistant to improve information accessibility at MUST.

CERTIFICATIONS

Mastering MLops	University of Kerala India	2026
Digital Entrepreneurship Education program	JA Africa	2024

EXPERIENCE

Current project

Magufuli Library AI – Mbeya University of Science and Technology

- Designing an AI-powered system to assist users in locating academic resources, answering library-related queries, and improving access to digital materials.
- Exploring retrieval-augmented generation techniques to improve contextual accuracy in responses.
- Aiming to enhance student research efficiency through automation and intelligent search

Industrial Practical Training – MUST IoT Club **July 2025 – September 2025**
Mbeya University of Science and Technology (MUST) | Mbeya, Tanzania

- Designed and deployed MUST AI Chatbot, an AI-powered assistant serving students, staff, and visitors by providing automated information about academic programs, admissions, campus services, and institutional procedures.
- Integrated natural language processing techniques to enhance contextual understanding and response relevance.
- Participated in the Digital Club Hackathon organized by Tanzania Communications Regulatory Authority (TCRA), where I developed an AI-based postal packet tracking system.
- The solution was shortlisted among the top 6 finalists, recognized for innovation and practical applicability.
- Collaborated with multidisciplinary teams to prototype scalable AI-driven systems.

Industrial Practical Training – Machine Learning & Artificial Intelligence **July 2024 – September 2024**
College of Information and Communication Technology

- Applied supervised and unsupervised learning techniques to structured datasets for predictive modeling tasks.
- Developed experimental ML pipelines including preprocessing, feature engineering, model training, and validation.
- Conducted performance evaluation using standard metrics such as accuracy, precision, recall, and F1-score.
- Documented technical workflows and presented findings to academic supervisors.

Mwalimu Nyerere Memorial Referral Hospital | Musoma, Mara **July 2023 – September 2023**

Data Analyst

- Performed data cleaning and statistical analysis on hospital service records to support operational reporting.
- Developed structured datasets from raw clinical and administrative records to improve reporting accuracy.
- Assisted in generating analytical summaries used in departmental decision-making.
- Improved data consistency and reporting turnaround through systematic preprocessing and validation

EDUCATION & QUALIFICATION

Mbeya University of Science and Technology (MUST) | Ongoing

Bachelor of Engineering in Data Science

Ihungo Boys Secondary School | 2019 – 2021

Advanced Level Secondary Education (PCM Combination)

HOBBIES & INTERESTS

- Continuous learning in emerging AI technologies
- Exploring new tools in machine learning and data science

LANGUAGES

English(Fluent) Swahili(fluent)

REFEREES

Available upon request